Task 17 – Disk Usage Monitoring Script with Email Notification via AWS SNS

Objective

Create a script on your instance which writes the disk usage to a file and send an email to specific mail.
Verify the recipient mail id in SNS console.
Configure Simple Notification Service (SNS) for delivering the mail notifications.
Verify the mail sent.

Requirements

- AWS EC2 instance running RHEL
- AWS CLI installed on the instance
- An IAM role attached to the instance with SNS publish permissions
- Amazon SNS topic and email subscription
- A bash script to generate the disk usage report and trigger the notification

Procedure

Step 1 – Launch the EC2 Instance

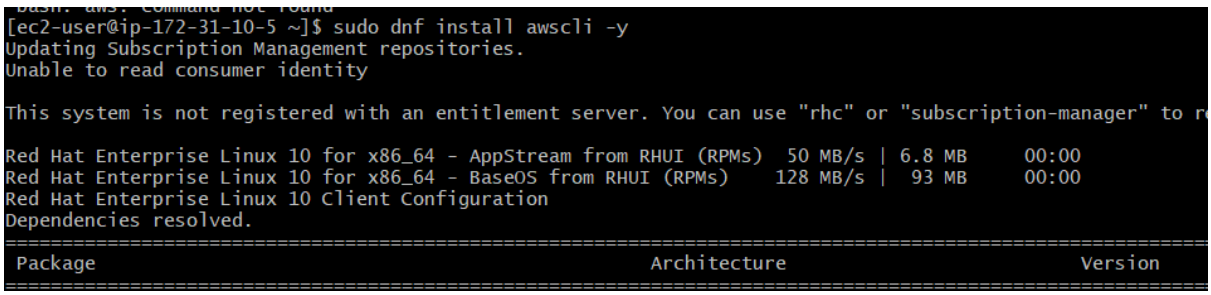
An EC2 instance running RHEL was launched to host and run the disk usage script.

Step 2 – Install the AWS CLI

The AWS CLI was checked and installed on the instance:

aws --version
sudo dnf install awscli -y

The installation was confirmed as version 2.33.0.



Step 3 – Create an IAM Role for SNS Access

Rather than using Access Keys, an IAM role was created and attached directly to the EC2 instance, allowing it to authenticate to AWS automatically without storing any credentials on the instance itself.

- IAM → Roles → Create role → Trusted entity: AWS service → Use case: EC2
- Policy attached: AmazonSNSFullAccess
- Role name: EC2-SNS-Publish-Role

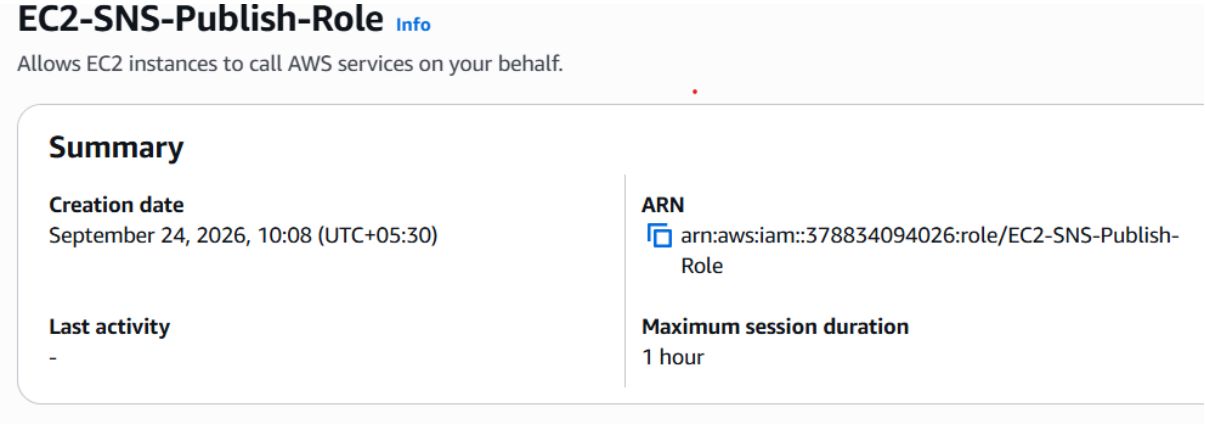
The role was then attached to the running instance via:

EC2 → Instance → Actions → Security → Modify IAM role

The role's credentials were verified using:

aws sts get-caller-identity

The response confirmed the instance was authenticating as the assumed role (EC2-SNS-Publish-Role), rather than requiring manually configured Access Keys.



Step 4 – Create the SNS Topic

An SNS topic was created to handle the disk usage notifications:

SNS → Topics → Create topic
Type: Standard
Name: disk-usage-alerts

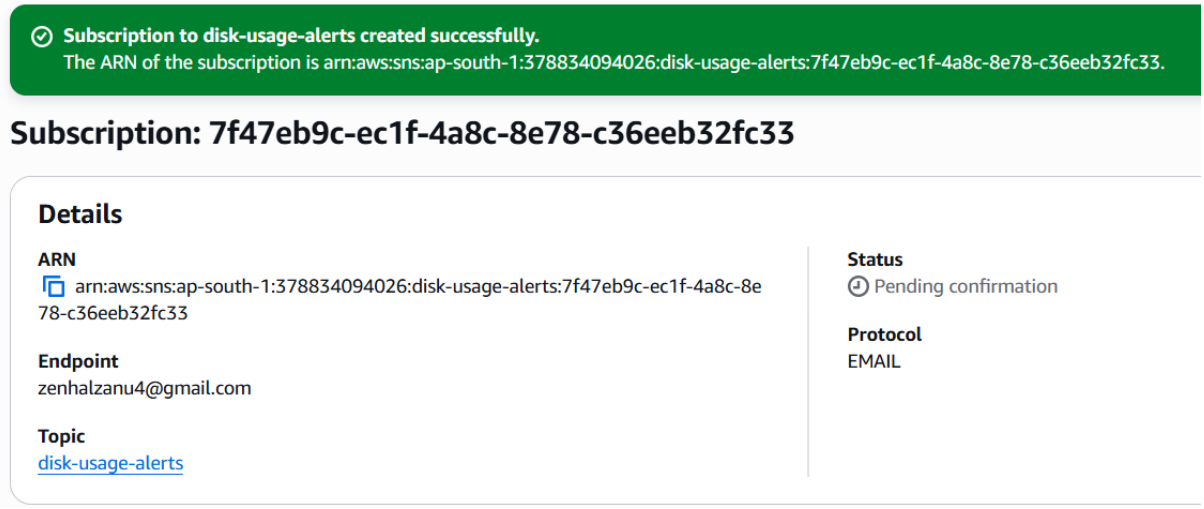


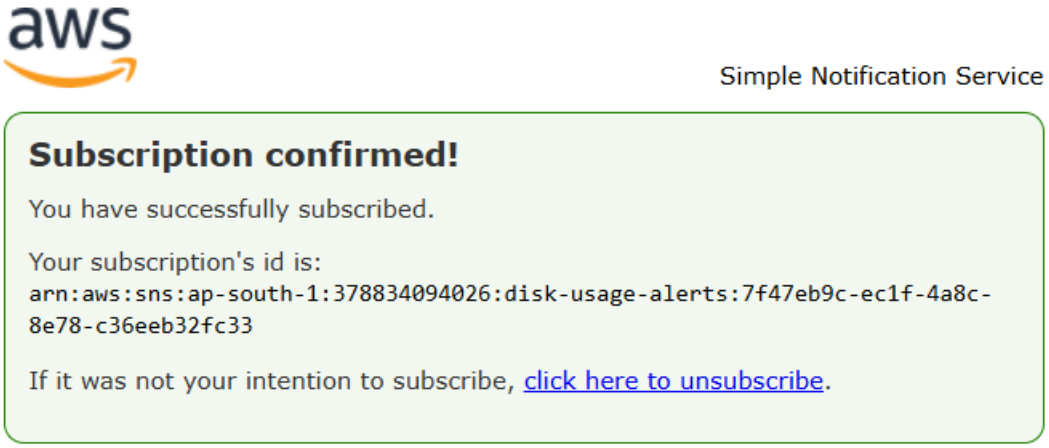
Step 5 – Create and Confirm the Email Subscription

An email subscription was created on the topic:

Create subscription
Protocol: Email
Endpoint: <recipient email address>

The recipient's inbox received a confirmation email from AWS Notifications. The subscription was confirmed by clicking "Confirm subscription" in that email, after which the subscription status changed from "Pending confirmation" to "Confirmed" in the SNS console. This step satisfied the requirement to verify the recipient email address in the SNS console.





Step 6 – Note the Topic ARN

The Topic ARN was copied from the SNS console for use in the script:

arn:aws:sns:ap-south-1:378834094026:disk-usage-alerts

Step 7 – Create the Script

A bash script was created on the EC2 instance to generate the disk usage report and publish it via SNS:

```
vi disk_usage.sh
bash
#!/bin/bash

OUTPUT_FILE="/home/ec2-user/disk_usage_report.txt"
SNS_TOPIC_ARN="arn:aws:sns:ap-south-1:378834094026:disk-usage-alerts"

echo "Disk Usage Report - $(date)" > "$OUTPUT_FILE"
echo "-----------------------------------" >> "$OUTPUT_FILE"
df -h >> "$OUTPUT_FILE"

aws sns publish \
  --topic-arn "$SNS_TOPIC_ARN" \
  --subject "Disk Usage Report - $(hostname)" \
  --message file://"$OUTPUT_FILE" \
  --region ap-south-1
```

The script was made executable:

```
chmod +x disk_usage.sh
```

Step 8 – Run the Script

```
./disk_usage.sh
```

The script executed successfully and returned a valid MessageId from AWS, confirming that SNS accepted the message for delivery:

```
json
{
   "MessageId": "15e85594-5ee6-5e1b-b632-36baf16b0293"
}
```

Step 9 – Verify the Output File

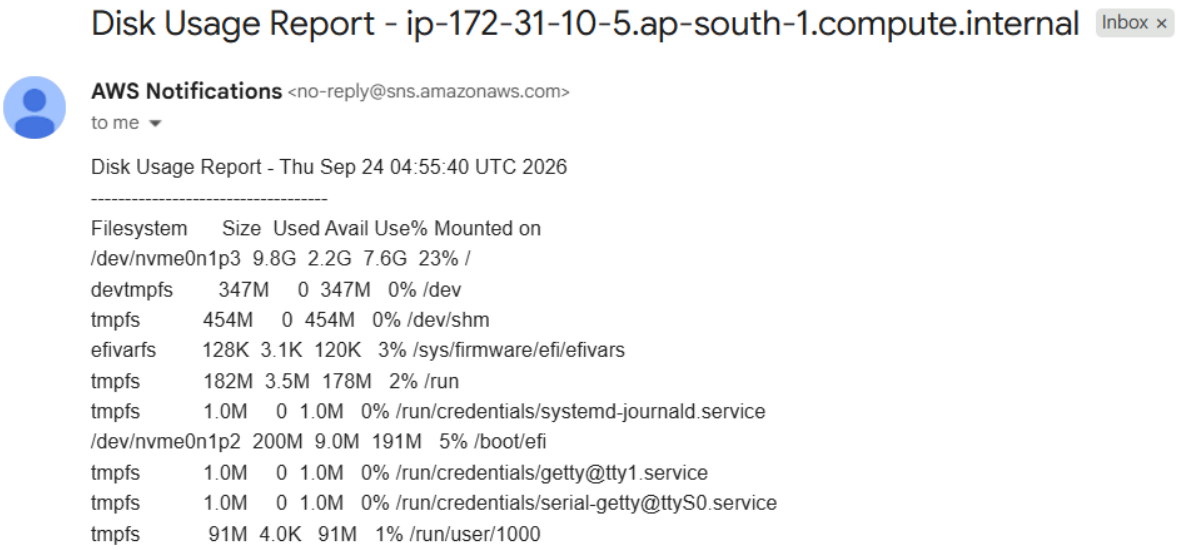
cat disk_usage_report.txt

The file correctly contained a timestamped disk usage report generated using df -h.

```
[ec2-user@ip-172-31-10-5 ~]$ cat disk_usage_report.txt
Disk Usage Report - Thu Sep 24 04:55:40 UTC 2026
-----------------------------------
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p3  9.8G  2.2G  7.6G  23% /
devtmpfs        347M     0  347M   0% /dev
tmpfs           454M     0  454M   0% /dev/shm
efivarfs        128K  3.1K  120K   3% /sys/firmware/efi/efivars
tmpfs           182M  3.5M  178M   2% /run
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2  200M  9.0M  191M   5% /boot/efi
tmpfs           1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           1.0M     0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
```

Step 10 – Verify the Email Notification

The recipient's inbox was checked and confirmed to have received an email from AWS Notifications (no-reply@sns.amazonaws.com), with the subject "Disk Usage Report - ip-172-31-10-5.ap-south-1.compute.internal" and a body matching the exact content of the generated disk usage report.



Verification

- The AWS CLI was successfully installed and verified on the EC2 instance.
- An IAM role with SNS publish permissions was created and attached to the instance, allowing authentication without manually configured Access Keys.
- An SNS topic was created to handle disk usage notifications.
- An email subscription was created and successfully confirmed, verifying the recipient's email address in the SNS console.
- A bash script was created to generate a disk usage report and publish it to the SNS topic.
- The script executed successfully, returning a valid SNS MessageId.
- The generated report file was verified to contain accurate, timestamped disk usage information.
- The notification email was successfully received in the recipient's inbox, confirming the correct configuration and delivery of the alert.

Conclusion

A bash script was successfully created on an EC2 instance to monitor disk usage and generate a report file. Amazon SNS was configured with a topic and an email subscription to deliver notifications, and the recipient's email address was verified directly through the SNS console. Using an IAM role attached to the instance, the script securely published the disk usage report to SNS without requiring manually stored AWS credentials. The email notification was successfully received, confirming that the entire monitoring and alerting pipeline — from disk usage check to email delivery — was functioning correctly end to end.